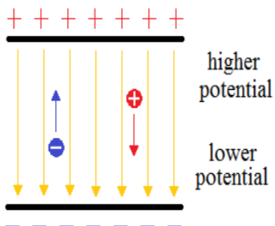


Spring 22

- nd answer
- Write at least three differences between electric potential energy? [2] CO1
 - Draw electric field lines from a dipole. Show directions of electric fields at three points in this field lines. [2] CO1
 - Direction of electric field along +Y axis. An electron is moving along +X axis. Does work done positive/negative; potential energy of electron increase/decrease? [4] CO3

Summer 21

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In figure, negative and positive charges are moving from lower to upper and upper to lower plates respectively. (i) Does negative/positive charge increase/decrease its electrical potential energy? (ii) Is the negative/positive work done by the electric field on negative/positive charge? [3] CO1
 - The direction of a uniform electric field is parallel to the +X axis. A dipole consists of $\pm q$ charges separated by a distance d . The dipole is placed with an angle $\theta = 60^\circ$ in the electric field. (i) Draw the arrangement of dipole and electric field. (ii) Show the direction of rotation of dipole in the electric field. [2] CO1

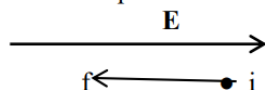
- Derive expression for electric field due to a dipole. Modify the expression when $z \gg d$. [4] CO2
 - OR Find the electrical potential due to line of charges.

Fall 20

- If two charges of equal magnitude are separated by a fixed distance r , what will be impact of electrostatic force on the charge? 1 CO1
 - What is electric potential energy? Is there any major difference between electric potential and electric potential energy? If any write down it. Graphically show the different electric potentials. 2 CO1
 - What is rms voltage? Draw a rms voltage graph if peak voltage is 40V. What is the main difference between ac and dc current? Mention at least two sources of AC and DC voltage source. 2 CO1
- Establish a relation between torque $\vec{\tau}$, electric dipole moment $\vec{p}$, and electric field $\vec{E}$. 2.5 CO2
 - Find out the electric field due to an electric dipole. 2.5 CO2
- Find out the electric potential due to a group of point charges. 2.5 CO2
 - Establish a relation between drift velocity and number density of electrons. 2.5 CO2

FALL 21

1. a) What are relations between electric field and line of forces [2] CO1
- b) In the figure, we move a proton from point i to point f in a uniform electric field. Is positive or negative work done by (a) the electric field and (b) our force? (c) Does the electric potential energy increase or decrease? (d) Does the proton move to a point of higher or lower electric potential? [3] CO1



- c) Draw some equipotential surfaces of a point charge. [1]

4. a) Derive expression for electric field for a charged ring. Modify the expression when $z \gg R$ and $z = 0$. [7] CO2
- b) Derive expression for electric field due an electric dipole [5]
5. a) Find the electrical potential for (i) Point charge and for (ii) a dipole in electric field. [6] CO2
- b) Find out electric potential at a point P, distance r from a charged wire of length L . [6]

FALL 20

1. (a) If two charges of equal magnitude are separated by a fixed distance r , what will be impact of electrostatic force on the charge? 1 CO1
- (b) What is electric potential energy? Is there any major difference between electric potential and electric potential energy? If any write down it. Graphically show the different electric potentials. 2 CO1
- (c) What is rms voltage? Draw a rms voltage graph if peak voltage is 40V. What is the main difference between ac and dc current? Mention at least two sources of AC and DC voltage source. 2 CO1

5. (a) Establish a relation between torque $\vec{\tau}$, electric dipole moment $\vec{p}$, and electric field $\vec{E}$. 2.5 CO2
- (b) Find out the electric field due to an electric dipole. 2.5 CO2
6. (a) Find out the electric potential due to a group of point charges. 2.5 CO2
- (b) Establish a relation between drift velocity and number density of electrons. 2.5 CO2

1. (a) The displacement of a Simple Harmonic Motion (SHM) is $y = A \sin(\omega t + \frac{\pi}{2})$. Find the expression of acceleration and graphically show the displacement and acceleration in graph. 2 CO1
- (b) For SHM, "the acceleration is proportional to the displacement". Justify the statement from Hook's law. 1 CO1
- (c) Draw the phase difference of two waves, when they are (i) in phase and (ii) out of phase. 1 CO1

4. (a) Suppose the displacement of a SHM is $x = -A \sin(\omega t + \delta)$. Find out the phase difference between velocity and displacement with necessary graph. 2 CO2
- (b) A spring block system is considered. Find out the instantaneous velocity $v(t)$ of a SHM. 2 CO2
5. (a) If you have inductor, capacitor and resistor, then draw a circuit comprising all. Obtain a differential equation for that circuit and also find ω . 2 CO2
- (b) Suppose, a travelling wave is moving from right hand side to the left. Now establish an equation for the progressive wave. 2 CO2

Fall 21

1. (a) If two charges of equal magnitude are separated by a fixed distance r , what will be impact of electrostatic force on the charge? 2 CO1
What is the graphical relation between charge and electrostatic force?
Can it be possible to apply the Coulomb's force in Navigation system? Justify your answer.
- (b) Is there any major difference between electric potential and electric potential energy? Justify your answer. How can you draw an electric potential graph varying in between -8V to +6V? 2 CO1
- (c) Suppose you imagine the angle between area vector and electric field vector is θ which produces a Gaussian surface. Now, how can you get the electric flux ϕ will be (i) zero, (ii) maximum, (iii) minimum, (iv) +ve flux, and (v) -ve flux. 2 CO1
5. (a) Find out the electric field due to an electric dipole. 4 CO2
- (b) Establish a relation between torque $\vec{\tau}$, electric dipole moment $\vec{p}$, and electric field $\vec{E}$. 4 CO2
6. (a) Find out the electric potential due to an electric dipole. 4 CO2
- (b) Suppose you imagine a Gaussian surface in the form of a closed cylinder surrounds a section of very large, uniformly charged cylindrical rod. Now applying the Gauss's law, find out the electric field. 4 CO2

FALL 22

1. (a) The displacement of a Simple Harmonic Motion (SHM) is $y = A \sin(\omega t + \frac{\pi}{2})$. Graphically show that the displacement and acceleration are out of phase to each other. 2 CO1
- (b) Why do $\omega^2 > \gamma^2 / 4$ oscillatory? Draw displacement vs. time graphs for (i) $\omega/\gamma = 5$ and (ii) $\omega/\gamma = 0.005$. 2 CO1
- (c) Graphically show that even though the potential and kinetic energies for Simple harmonic motion (SHM) vary with time, the total energy remains constant. 2 CO1
- (d) Derive differential equation for Simple pendulum. Find out expression for frequency of oscillation. 4 CO2
- (e) Derive the differential equation for RLC circuit and find out the condition for its oscillatory behavior. How the oscillation of the RLC circuit becomes that of an LC circuit? 4 CO2

Spring 23

1. (a) What is equipotential surface? 2 CO1
 (b) Identify if the work done is positive or negative when the $+q$ charge moves from point A to point B (figure 01)? Does the potential energy increase or decrease in figure 01? 2 CO1

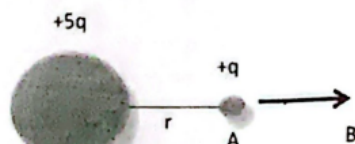


Figure 01

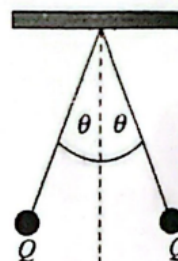


Figure 02

- (c) Two positively charged balls are suspended as shown in figure 02. Draw the figure with appropriate electric field lines. 2 CO1

5. (a) A dipole with charge $+q$ and $-q$ are separated at a distance " d " and lie along the dipole axis (z axis). P is a point along the axis and it is at a distance " z " from the center of the dipole. Calculate the electric field at the point P due to the electric dipole. 4 CO2
 (b) Show that, a potential energy U is associated with the orientation of the dipole moment in the field, as given by a dot product: 4 CO2

$$U = -\vec{p} \cdot \vec{E}$$

Where $\vec{E}$ is the external electric field and $\vec{p}$ is the dipole moment.

$$\begin{aligned} dW &= \tau d\theta \\ dW &= Fr \sin \theta / r d\theta \\ dW &= qE \sin \theta \end{aligned}$$

6. (a) Show that, the electric potential due to a single charged particle at a distance r from that charged particle is, 4 CO2

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

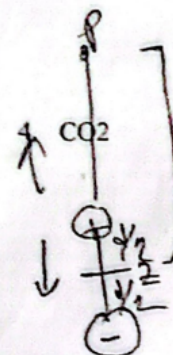
Where V has the same sign as q .

Write down the formula for electric potential due to a group of point charged particles.

- (b) At a distance r from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is ,

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

Where $r \gg d$; the angle θ lies between the dipole moment vector and a line extending from the dipole midpoint to the point of measurement.



Spring 21

1. (a) Justify that the dielectrics should not conduct current to construct a capacitor. 2 CO1
 (b) Justify that the resistance can vary for the same material with a certain change of parameters, but resistivity does not. 2 CO1
 (c) To have maximum use of electrical power what combination of four bulbs should you use? Justify your answer. 2 CO1
 (d) Define work function and threshold frequency. 2 CO1

5. (a) Find out the electric potential due to an electric dipole. Rewrite the case if the electric charge is continuous. 4 CO2
(b) Establish a relation between drift velocity and number density of electrons as well as the relation between conductivity and current density. 4 CO2
6. (a) Find out the equivalent capacitance, C_{eq} for series and parallel combinations of four capacitors with each has the half capacitance of the previous one. 4 CO2
(b) Suppose the voltage varies sinusoidally with time as $V = V_0 \cos 2\pi f t$. Find out the average electric power $\bar{P}$ in a typical home. Also compare with necessary figures for average power and electric power. Draw rms voltage figure for above mentioned dynamic equation of the voltage equation. 4 CO2